

Classification of Internal and External Distractions in an Educational VR Environment Using Multimodal Features

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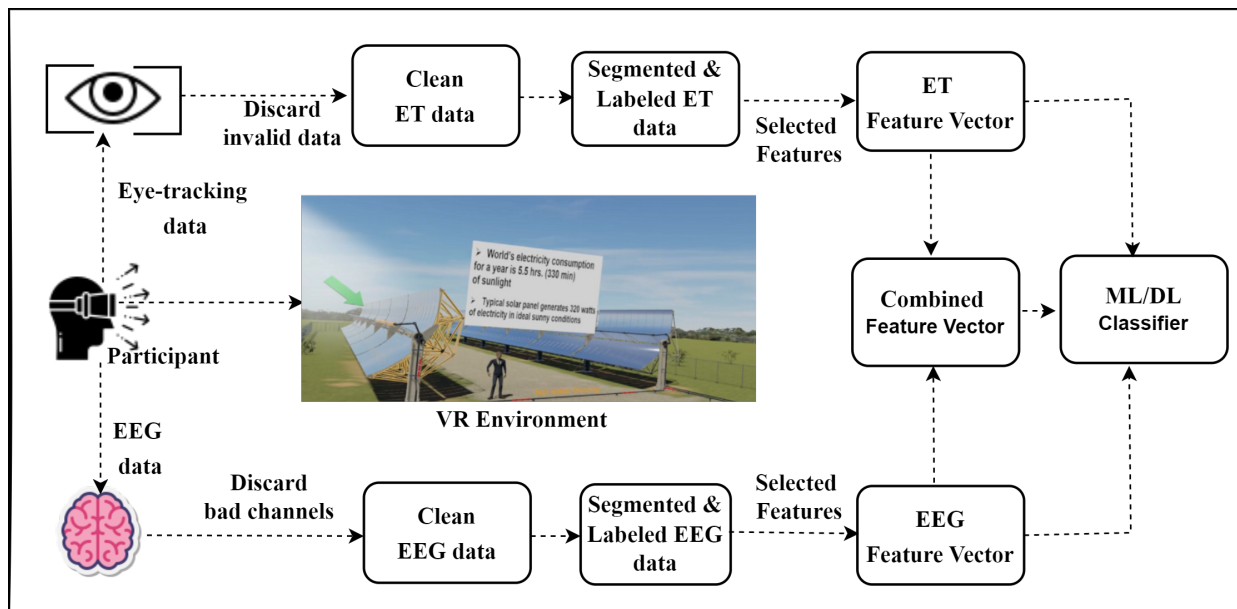


Fig. 1: EEG and eye-tracking data preprocessing pipeline to extract features for classification of internal and external distraction.

Abstract—Virtual reality (VR) can potentially enhance student engagement and memory retention in the classroom. However, distraction among participants in a VR-based classroom is a significant concern. Several factors, including mind wandering, external noise, stress, etc., can cause students to become internally and/or externally distracted while learning. To detect distractions, single or multi-modal features can be used. A single modality is found to be insufficient to detect both internal and external distractions, mainly because of individual variability. In this work, we investigated multi-modal features: eye tracking and EEG data, to classify the internal and external distractions in an educational VR environment. We set up our educational VR environment and equipped it for multi-modal data collection. We implemented different machine learning (ML) methods, including k-nearest-neighbors (kNN), Random Forest (RF), one-dimensional convolutional neural network - long short-term memory (1D-CNN-LSTM), and two-dimensional convolutional neural networks (2D-CNN) to classify participants' internal and external distraction states using the multi-modal features. We performed cross-subject, cross-session, and gender-based grouping tests to evaluate our models. We found that the RF classifier achieves the highest accuracy over 83% in the cross-subject test, around 68% to 78% in the cross-session test, and around 90% in the gender-based grouping test compared to other models. SHAP analysis of the extracted features illustrated greater contributions from the occipital and prefrontal regions of the brain, as well as gaze angle, gaze origin, and head rotation features from the eye tracking data.

Index Terms—Machine Learning, Human-centered computing, EEG, Eye-tracking

1 INTRODUCTION

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VR-based modern classroom learning aims to provide vivid visual learning experiences to the students for better remembrance. It can also improve teaching skills [25]. A significant problem in VR-based learning is the distraction among the participants. Distractions in VR may arise from stress, mind wandering, task-irrelevant thoughts, external alerts/noise, and other factors like dizziness, sickness, etc. In a physical classroom setting, teachers can gauge students' engagement and behavior through multiple cues, including body language, eye contact, and facial expressions. Hence, the attention levels of the students can roughly be estimated. However, in VR environments, the teachers do not have direct visibility of the students, leading to a reduced awareness of their activities, which results in difficulty in measuring the distraction levels of the students. According to numerous studies in psychology, cognitive science, and neuroscience [15, 17, 34], distractions significantly degrade the learning outcomes, students have less active brains toward their learning, and their performance of knowledge gain is less

efficient. Moreover, distractions can increase stress and anxiety, hinder student's ability to think critically and analyze information, etc. Therefore, distraction detection among the students in VR-based classrooms is promising to assist the teachers in running better and more effective learning.

Distractions in VR can be categorized into external (environmental factors) and internal (thoughts in mind). Distraction happens due to a lack of attention towards a relevant task. Internal distractions in students happen due to irrelevant thoughts in their minds and focus on unrelated tasks. In contrast, external distractions could occur due to many external noises/sounds and visuals around the VR surroundings. Distraction can be classified using non-physiological and/or physiological signals obtained from the participants. Teachers may try to understand a distracted student directly from his/her non-physiological features, e.g., facial expression, speech, gesture, etc. However, analyzing distractions using physiological features, like pupillary diameter, eye gaze, electroencephalograph (EEG), electrocardiogram (ECG), etc., requires computer-based efficient algorithms, e.g., ML methods. Distraction detection using physiological data and ML methods is more practical compared to manual assessment using non-physiological data, particularly in VR environments. Therefore, the ML methods could reliably classify and detect students' distractions, and significantly improve VR-based education.

Among different physiological modalities, eye tracking (ET) data has widely been used for attention/distraction analysis in VR applications [3]. ET devices capture the gaze positions toward the virtual objects in VR environments, from which the distraction level can be estimated. However, such techniques for distraction detection using ET features (such as distance of objects) are specific to a virtual environment with specific virtual tasks [3]. Apart from the ET modality, EEG is another popular modality used in attention/distraction detection in VR. Many studies found that brain signals from the EEG device get affected during inattentive tasks and, hence, can be reliably used for attention/distraction detection [43]. EEG data is, therefore, not specific to virtual tasks, rather, any distracted action of a participant can make a footprint in the EEG data. While eye-tracking is an external physiological feature, combining it with an internal physiological feature, like the EEG data, the accuracy and reliability of the distractions detection algorithm can significantly be improved. Additionally, any single modality cannot accurately classify both internal and external distractions [19]. For example, a student may be thinking about something else while watching the educational material. In such a scenario, eye-tracking data as an external physiological feature can only reliably detect external distractions and will perform poorly in detecting internal distractions. Therefore, it is essential to combine these internal and external physiological data, e.g., ET and EEG, to perform a more accurate and reliable classification of both internal and external distractions.

In this study, we examined the effects of using the EEG and eye-gaze data to classify both internal and external distractions using ML techniques. To accomplish this, we developed two educational virtual reality (EVR) environments (refer to Figure 2) and gathered EEG and eye-gaze data from 27 subjects. The resulting dataset was utilized to train our models, including 1D CNN-LSTM, 2D CNN, RF, and kNN, to classify distractions. We considered two classes: internal distraction (ID), and external distraction (ED). We conducted cross-subject and cross-session testing to evaluate our model and report the results. Figure 1 shows a pre-processing pipeline of multimodal data. In addition, we analyzed the potential effects of the participant's gender on classifying internal and external distractions, which will be elaborated more in the following. We report the results of our analysis on both collected sensor data and user experience data. Our obtained results contribute new evidence for ongoing EVR research and can inspire future work.

The contributions of this study are as follows:

- We designed two EVR environments for distraction experiments and a novel approach to detecting internal and external distraction using EEG and eye-tracking data collected from 27 participants when they performed internal and external distraction tasks.
- We evaluated the performance of four ML models (1D CNN-

LSTM, 2D CNN, RF, and kNN) on the three different feature sets (raw, statistical, and image features) by combining the EEG and eye-gaze data using an early feature fusion approach to classify participants' internal and external distraction states. In addition, we found that gender-based data grouping significantly enhanced the classification accuracy of internal and external distractions.

2 RELATED WORK

In this section, we present relevant research on multimodal data analysis and distraction detection using physiological data.

2.1 Distractions in EVR

In an educational program, various activities can divert students' attention, whether it is in VR or not. According to psychological research, many students are reported to use their phones during class to use social media, shop online, explore the internet, or be involved in non-academic activities [31]. Multitasking reduces the attention level and also downgrades the ability of cognitive processing, making it difficult to remember information [13,39]. When a broad array of visual stimuli is presented to the students in a VR environment, it becomes harder to focus on the learning goal [14]. Bozkir et al. [11] found that students sitting at the front of a virtual reality classroom paid more attention to the virtual instructor and lecture material than those sitting at the back. However, Han et al. [17] investigated that students near the back of the classroom were more attentive to the non-essential elements. Additionally, Bozkir et al. [11] reported that VR lectures over 45 minutes could produce severe cybersickness, resulting in a considerable loss of attention. After studying different factors that affect the attention level, in this research, we focused on limiting our VR experiment duration (≈ 22 min) to minimize cybersickness and distraction. We also placed the participants in the same sitting position, and fewer but relevant visual stimuli were presented to them. Recently, gender differences have been discussed in different educational settings. For instance, Barmaki et al. [7] found that male students reported more positive learning experiences than female students for anatomy education. Prior studies have also demonstrated that women perform better than men on cognitive tests regarding their ability to comprehend information [36,41]. However, limited research has been done on the effects of gender on distractions. This motivated us to perform a gender-based group test classifying internal and external distractions.

2.2 Detecting Distractions using EEG and Gaze Data

Several studies have developed ML models using multimodal data to analyze mind wandering and attention/distraction in different contexts. Vortmann et al. [45] utilized multimodal EEG and eye-tracking data to train convolutional neural networks (CNN) for internally vs. externally directed attention detection in a non-VR application. In a similar non-VR work, Zheng et al. [49] utilized EEG and eye gaze data for emotion recognition using an SVM classifier. Rodrigue et al. [39] detected divided attention in a non-VR reading application using EEG and eye gaze data by applying a customized ML model. Asish et al. [1,3] used eye gaze data to determine the external distractions using deep learning models. Another work by Asish et al. [2], used EEG data to detect internal distractions using machine learning models. They also used a multimodal approach combining ET and EEG data collected from 10 participants with limited settings of VR design [4]. However, detailed data analysis was missing and the result was not conclusive. All the mentioned scientific works have used EEG and/or eye-tracking to measure attention/distraction in different applications, but they have not fully examined internal and external distractions in the same EVR environment. Moreover, none of these works studied multimodal data to classify both internal and external distractions in EVR. This has motivated us to conduct a study in an EVR to detect internal and external distractions using multimodal features. Our research question was - to what extent could we detect internal and external distractions using EEG, eye gaze, or a combination of both data types when the students get lost in their thoughts or when external disturbances disrupt their attention?



Fig. 2: EVR environment: (a) An avatar explaining the nucleus cell during the practice session. (b) An avatar describing a solar panel during the actual VR experiment.

With the research question unfolding, we are motivated to aid students in VR-based classroom learning. We, therefore, conducted our experiments in an EVR settings. We designed our VR environment containing an avatar, audio, text slides, and animations. We simulated common internal and external distractions to make the VR class more realistic. We collected EEG and gaze data to classify participants' internal and external distraction states using different ML models.

3 EDUCATIONAL VR (EVR) ENVIRONMENT

We used a Virtual Energy Center [9, 10] to simulate internal and external distractions in our VR environment from virtual solar field trips (see Figure 2(b)). This VR environment showed how different components in a power generation plant work. The VR session delivered multiple learning components (an avatar, animations, audio, and slides) played simultaneously. Prior research suggests that using an avatar can help the students to better learn [26]. The teacher's avatar can point and play animations to draw attention to the topic in our environment. In the past, such animations have been used to illustrate the inner workings of a given object [37]. Baceviciute et al. [5] found that reading is more effective than listening. Yet, the audio in that research was not paired with any visuals (such as text, avatars, or animations). Our research used text slides to capture important component phrases and mathematical equations. Prior research showed that text slides helped students retain knowledge, especially for mathematical topics and equations that are difficult to understand verbally. However, a study by [29] indicated that while multimedia slides increased users' interest, they did not improve the learning outcomes. Therefore, in our VR environment, all components were synced to illustrate the subject, and relevant solar field components were highlighted to catch students' attention. We believe combining these teaching components will increase student engagement. In addition, we created a biology-focused VR environment for learning cell structure (Figure 2(a)) that participants utilized as a practice session before the main experimental tasks. In this environment, an avatar explains about a nucleus cell and its components using gestures and 3D visualizations of the components involved. This presentation was 1 minute 42 seconds long.

4 METHODOLOGY

4.1 Method Overview

In order to train and evaluate various machine learning models, we gathered EEG and gaze data from our VR environment, as will be described below. Once participants gave their informed consent, the experimental activities were explained to them, and they had a chance to practice the tasks using the biology environment (see Figure 2(a)). The participants next started the experimental tasks, which required them to focus on a VR presentation regarding solar panels (see Figure 2(b)). In this experiment, we had two phases with identical teaching material, each with four sessions (duration ranging from 1.5 to 4 minutes). Each session covers a small topic related to solar panels. Our experimental workflow is summarized in Figure 4.

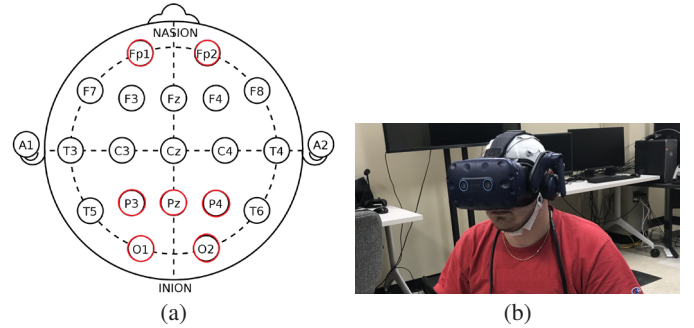


Fig. 3: VR device and EEG equipment used in our study. (a) The positions of electrodes in the International 10-20 system used for EEG recording. The red circles indicate the EEG channels selected for our analysis. Wikipedia provided this free image license [48]. (b) User Setup for the experiment with the OpenBCI all-in-one electrode cap and the HTC Vive Pro Eye headset.

4.2 Participants and Apparatus

We collected data from 27 participants (17 male and 10 female) from the university population. Their ages ranged from 18 to 30 (mean age was 21.4 and standard deviation was 2.91) and 18 of them had prior VR device experience. Their races include White American (12), Asian (10), Black or African American (3), Native Hawaiian (1), or other Pacific Islanders (1). 12 out of the 27 participants were international students. The whole experiment took about 40 to 60 minutes, including the time it took to get their permission, set up the eye tracker, and talk briefly about their VR experience. The total duration of this VR experiment of two phases took around 22 minutes. Two participants came back after five weeks for two sessions to evaluate our pre-trained ML model from a different day.

A desktop machine (Core i9 11900F, Microsoft Windows 10 Pro, NVIDIA GeForce RTX 3080 Ti, 64 GB RAM) was used to run the Experiment and implement the VR tasks in Unity 3D v2018.2.21f1. We collected EEG data using the OpenBCI all-in-one electrode cap kit with a sampling rate of 125 Hz, and eye gaze data were collected at 120 Hz using the HTC Vive Pro Eye (see Figure 3 for user setup (b)). For our ML scripts, we relied on Python (version 3.8.8) and its scikit-learn, TensorFlow, and Keras libraries.

4.3 Experiment Design

In this experiment, we considered both internal and external tasks to create distractions since students are distracted due to external reasons or internal thoughts, according to real-world classroom distraction research. We divide our experiment into two phases: Phase I with internal distraction tasks and Phase II with external distraction tasks. Each phase comprised four sessions with two exam questions based on the content presented.

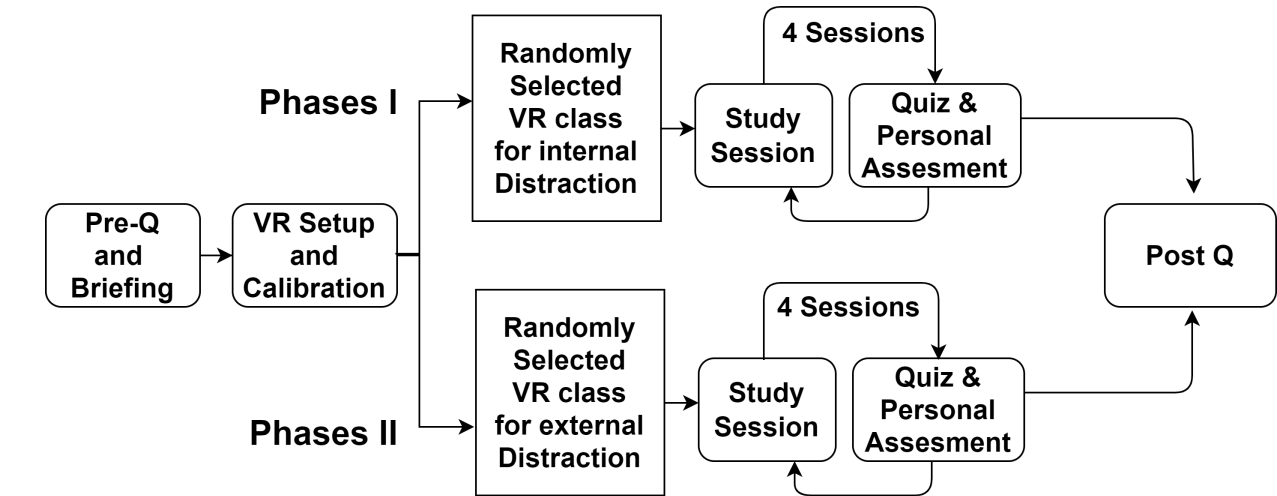


Fig. 4: Experiment Workflow

Table 1: Internal distraction tasks adapted from [4] were applied at random in the VR scene, and a doorbell sound was used to alert participants to begin the task.

Internal task to create distractions	
1	Count how many times the avatar moves his hand.
2	Count the total number of words starting with 'S'.
3	Think about the activities for the upcoming weekend.
4	Think about the activities during the last weekend.

Table 2: Dialogues used to divert attention away from a related task to create a distraction (borrowed from [3])

Dialogues to divert focus/attention	
1	Think about your last conversation with your family.
2	Think about a current work challenge you are facing.
3	Think about a bird you saw recently.
4	Think about anything that crosses your mind.

Table 3: Post-Session Questionnaires completed after each session. The moderator typed the answers while participants answered these questions verbally.

Post-Session Questionnaire	
Please summarize what you did after the doorbell sound in this session.	Text type answer
Did you feel distracted during the task?	yes/no

4.3.1 Phase I with Internal Distractions

Internal distractions occur due to many factors such as internal stress, mind-wandering, daydreaming, illness, etc. Simulating internal distractions in an experimental VR setup is challenging due to individual variability. Since we used an EVR session to distract attendees, we focused on relevant internal tasks. While watching the VR lesson, these internal tasks were secondary. In Phase I, we used four internal tasks (see Table 1), which appear randomly about 50 to 60 seconds after the session begins. A doorbell sound rang for 1 second to read the instruction of an internal distraction task picked randomly from Table1. According to instructions, participants had to think about internal distraction tasks for 15–20 seconds so that they could tell 2–3 sentences from their internal thoughts. The remaining two questions were counting avatar hand movements and the number of words starting with “s”. In our preliminary research, participants often missed the visual instructions to start the internal distraction task. Since internal distraction is hard to simulate in VR for participants, we decided to alert them using a one-second doorbell sound to start the internal distraction task and control the experiment. After collecting data, we carefully discarded the data segments for external stimuli used for starting internal tasks. Then, we used participants’ questionnaire responses and manually inspected internal distraction segments to consider them in our analysis. The remaining data segments were discarded. However, we did not consider when the participants were naturally distracted (other than our simulated distractions) during data collection since controlling participants’ natural distractions in an EVR session is not feasible.

4.3.2 Phase II with External Distractions

In Phase II, we provided five external distractions for the participants while watching VR sessions: one graphical glitch and four external sounds. We played the external sounds along with the audio for the VR session through the VR headset. The four external sound distractions were (1) a pre-recorded mobile ringtone, (2) an external audio conversation between two people lasting for 10 seconds, (3) random

dialogue (selected from Table 2) played to distract students from the teaching content, and (4) The door’s opening and closing sounds resembled a classroom door. External conversation and dialogue are slightly different. While dialogue involves selecting a single, noiseless voice randomly picked from the table, conversation involves the noisy sounds of two people talking to each other. Prior research found that dialogues can distract for 15 seconds [23]. Our fifth distraction involved a simulated camera glitch (scan line artifacts) lasting for 10 seconds, which was designed to distort the virtual reality screen. These glitches may distract participants [50]. For each session in Phase II, these distractions appear randomly every 40 to 50 seconds.

4.3.3 Experiment Questionnaires

Our experiment used three questionnaires adapted from [1, 3]: one was administered prior to the experiment, one after each session, and one given at the end of the experiment. The pre-questionnaire gathered demographic information and inquired about respondents’ distractibility using the cognitive failure questionnaire (see to appendix Table 8). This was done to assess the level of distraction experienced during routine activities in the past six months [46]. The questions were answered using a 5-point Likert scale by the participants. Participants completed a questionnaire (Table 3) after each session to report internal or external distractions. This helps with data labeling later for the ML model. After all sessions, participants completed a post-questionnaire (see appendix Table 9) to evaluate overall impression of VR.

4.4 Data Collection Process

Participants were briefed on the study's procedures and told they would watch EVR content in many sessions lasting 1.5 to 4 minutes. They must report their distraction level after each session. Participants sat two meters from the facilitator after signing consent. Participants completed the pre-questionnaire. After that, they were asked to wear an EEG cap and an HTC VIVE Pro Eye VR headset. After setting up the VR headset and EEG device, participants were also asked to avoid excessive hand/leg movements because muscle activity in the legs/arms could affect the EEG data. Participants practiced internal distraction activities in the biology environment illustrated in Figure 2(a). The first practice session was watching biology lessons in VR. In the second practice session, they counted the avatar's hand, head, and leg movements that randomly appeared after a doorbell sound while watching the animal cell structure. No data was obtained during practice. After the practice session, we calibrated the eye tracker and checked the EEG cap. The participants then completed Phases I and II, each with four sessions in random sequence. After each session, they answered a quiz (for example, "How many hours of sunlight are required to power the entire world for a year?") and post-session questions (Table 3). In phase I, participants think about the internal distraction task for 15–20 seconds, and express 2–3 sentences based on their thoughts at the end of each session. We labeled it an internal distraction if they said the quiz question could not be answered accurately and agreed to internal distraction. Otherwise, we ignored it. In phase II, participants went through the four VR sessions along with external distractions. At the end of each session, if they have inaccurate quiz answers and feel distracted while viewing the session, we consider the data external distraction. After both study phases, the participants completed a post-questionnaire (see appendix Table 9). We also asked if the VR experience was distracting and if it helped them remember the information. Our experimental approach is shown in Figure 4.

We recorded both eye tracking (ET) and EEG data simultaneously during the two phases of the experiment as described below:

- **Eye-Tracking Data:** The recorded raw ET data comprises several features, such as eye openness, timestamps, eye wideness, pupil diameter, gaze direction, and gaze position. Additionally, we collected HTC Vive's reported gaze origin value, consisting of a 3D vector for each eye. Each data frame returned by the eye-tracking API was marked with a validity flag, and the sampling rate for gaze was 120 Hz. We also calculated gaze features, including head gaze angle, eye-gaze angles, head orientation, and distance features. Two eye-gaze angles were computed: one with respect to the highlighted object and another with respect to the teacher's avatar. The eye-gaze angle, relative to the highlighted object, was calculated as the angle between the gaze direction and the vector connecting the eye origin and the highlighted object. Similarly, the eye-gaze angle relative to the teacher avatar was computed as the angle between the gaze direction and the vector connecting the participant location (eye origin) and the teacher avatar. The angle between the head-gaze direction and the vector that connected the participant's location to the highlighted object was used to compute the head-gaze angle with respect to the object.
- **EEG Data:** During the sessions, raw EEG data from the participant's frontal, central, occipital, and parietal lobes was captured using the OpenBCI all-in-one electrode cap kit. This kit comprised 16 channels, and for each participant, we positioned the electrodes according to the international 10-20 system [18]. While recording the EEG data, we applied a notch filter at the power frequency of 60 Hz, and the bandpass filter was set to 1 to 50 Hz. To match the eye-tracker sampling rate, we down-sampled 125 Hz EEG data to 120 Hz.

4.5 Data Pre-Processing

It is necessary to pre-process the raw data from the experiment to create the input and output for building ML/DL models. The EEG and the ET data were pre-processed using simple methods to eliminate the noise.

By applying a z-score analysis to the data, we could eliminate noisy data that exceeded three standard deviations from the mean.

4.5.1 ET Data Pre-Processing

We eliminated all data values that the eye-tracking API had flagged as being invalid. However, some values were missing for some calculated features, like distance and eye/head gaze angles, and were logged as "NaN." All the missing and "NaN" values (less than 1% of data) have been replaced as part of our data cleaning process with the last known data values from the same feature.

4.5.2 EEG Data Pre-Processing

In this study, we used a 16-channel EEG cap. However, EEG data contains lots of artifacts that are hard to process and many of these channels are not related to attention/distraction. Thus, we had just seven channels for recording data for each participant with electrode locations (see Figure 3(a)) of FP1, FP2, P3, P4, Pz, O1 and O2 according to 10-20 system [18]. We selected these channels because the occipital lobe (O1, O2) is responsible for vision processing [30], the parietal lobe (P3, P4, Pz) provides information about attentional demands [22], and the prefrontal cortex (FP1, FP2) is in charge of decision making, cognitive state, and problem-solving [33]. During EEG recording, we used the OpenBCI tool's settings to apply a notch filter at the power frequency of 60 Hz, a bandpass filter of 1 to 50 Hz, and smoothing turned on. In this work, we collected raw EEG data. Lots of artifacts due to eye movements, hand/leg movements, and heart rhythms get added to EEG data due to the nature of the measurement process [44]. These artifacts are very challenging to remove from the EEG data. To clean the EEG data, we used the independent component analysis (ICA) to remove artifacts using the MNE-Python library. Moreover, we paid attention to the experiment setting process to minimize the occurrence of these artifacts (asked the user to reduce leg/arm motion as much as possible) instead of removing all artifacts using independent component analysis (ICA) [20, 44]. We used the least number of channels (only seven) for our classification task so that artifacts can't significantly impact our analysis results. The EEG data has been proven to be suitable for classification tasks [42] in the past.

4.5.3 Data Standardization

The last stage of the pre-processing step was to perform data standardization to scale the data since it is a required step for the deep learning models [16]. We standardized the EEG (per channel) and the eye-gaze data with the following equation: $X_t = (X_t - \mu) / \sigma$, where μ indicates the mean value (computed from all subjects) of the corresponding data, and σ represents the standard deviation. A data standardizer was used for the training and testing data separately. Standardization ensures that features with larger scales don't dominate the modeling process, preventing potential bias in the results.

4.6 Data Fusion

There are three techniques used for multimodal data fusion: (1) Early fusion or fusion at the data level, (2) Decision-level fusion or late fusion, and (3) intermediate fusion [24]. Ramachandram et al. [38] argue that there is no compelling evidence that late fusion is superior to early fusion. When the sampling rate is similar for both modalities, the early fusion is the most common in ML literature [6] because of the limited number of models (one) needed. Thus, we used an early fusion approach in our experiment since we have similar sampling rates for the EEG and the ET data, 125 Hz and 120 Hz respectively. Two feature sets were produced, one for each modality. The input from both modalities must be temporally compatible to be combined using the timestamp value because we considered early feature fusion methods. Thus, we down-sampled the EEG data to 120 Hz to match the eye-tracker data sampling rate. The total number of data samples was 347,640.

4.7 Data Labeling

The study examined how long simulated distractions lasted and how engaging was the educational VR experience. Initially, we considered

different distraction window lengths of 10, 15, or 30 seconds. We conducted a pilot study comparing 20-second and 30-second windows for internal distraction detection. 20-second windows outperformed 30-second windows. Thus, this experiment used the 20-second window. For the internal tasks, if the subject indicated being distracted in the post-session questionnaire, we clipped and categorized the data as Internally-Distracted (ID) for the 20 seconds starting 5 seconds after the doorbell sound (see Figure 5). It might be possible that the doorbell sound (one-second duration) creates external distraction; thus, we discard the data for 5 seconds from the start time of the doorbell sound. For external tasks, we sliced and labeled data as Externally-Distracted (ED) if participants said they were distracted during the external distraction event (see Figure 6). This resulted in a two-class (binary) classification task: Internally Distracted (ID) versus Externally Distracted (ED). We discovered that the quantity of data points related to each class of distraction (ID and ED labels) varied slightly. The data was more skewed in favor of the ED class. To avoid the bias, we used Synthetic Minority Over-sampling (SMOTE) [12] to "SMOTE" the minority class and under-sample the majority class in the training data to prevent bias and overfitting.

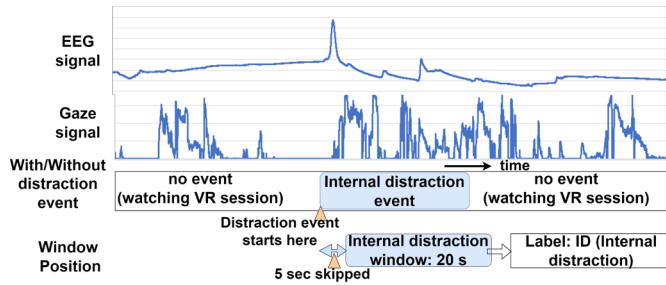


Fig. 5: Windowing and labeling of EEG and gaze data sequences for internal distraction. More details are provided in the Data Labeling section.

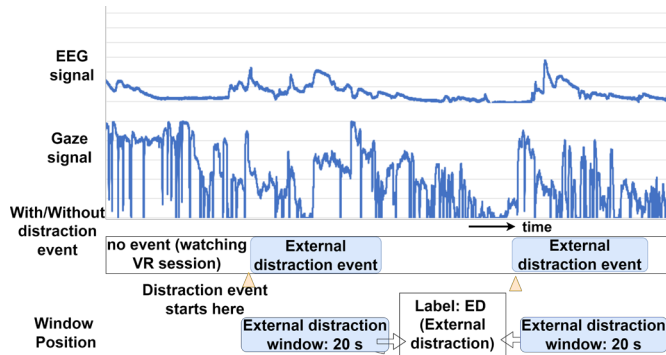


Fig. 6: Windowing and labeling of EEG and gaze data sequences for external distraction. More details are provided in the Data Labeling section.

4.8 Feature Selection

A study found that Random Forest (RF) can generate multivariate feature significance scores from high-dimensional or time series data [32]. Important scores are the result of implicit feature selection and might indicate feature relevance. We used RF to determine which properties from both modalities were most essential in our dataset. Figure 7 shows the scores for all 28 features. EEG features, pupil diameter, openness, gaze angle, and 3D gaze origin are highly correlated. Pupil diameter and eye openness were used to estimate drowsiness based on past research [3]. However, drowsiness and head rotation quaternion values (w) have low scores. These two less important features did not increase classification performance; hence, they were disregarded.

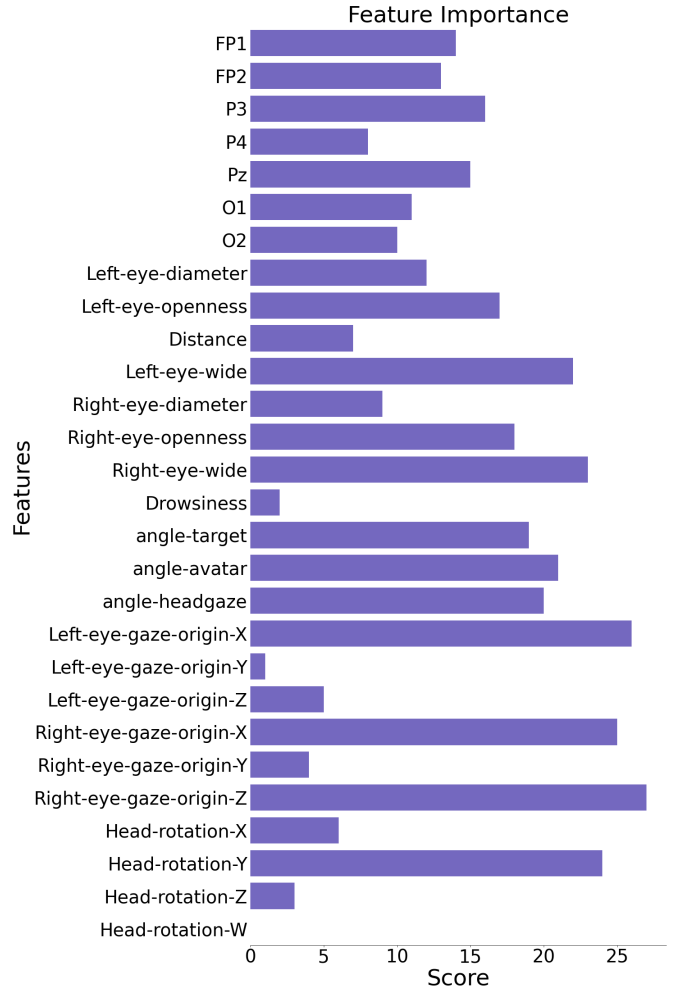


Fig. 7: Feature Importance Scores.

We experimented with combinations of the 26 important features and achieved the best results with a multimodal features set.

Based on past research [3,45], this study included raw, statistical, and image features for the four classifiers. We extracted statistical features from 1, 2, and 3-second sliding windows. Out of these, the one-second window performed better and we reported the results based on this. First, we used raw features for the 1D CNN-LSTM model introduced in [3]. Second, we turned our tabular raw data into image (26x26) data to train and test the 2D CNN model. After fusing data, we extracted signal characteristics using a 1-second sliding window without overlap. The features used by the kNN and RF models include statistical features (mean, standard deviation, min, max) for each column of the current window. We extracted $26 \times 4 = 104$ features for each window and added all window features to the dataset for the kNN and RF models.

4.9 Classification Models

We looked at four different ML models based on previous research: 1D CNN-LSTM, 2D CNN, RF, and kNN. The 2D CNN was chosen since it provides the best results for multimodal data classification [45], and the other three models were adopted from [3,8] since they worked well on the eye-tracking and EEG data. These models are described below:

1D CNN-LSTM: We used 1D CNN-LSTM model introduced by prior research [3,40]. While the convolutional neural network (CNN) layers are used to extract features from EEG and eye-gaze data, the long short-term memory (LSTM) layer is utilized to learn temporal features. The model uses two 1D-CNN with the parameters kernel=4, stride=2, and Maxpool=2. ReLu was used as the activation function. The output of the max pool layer follows one LSTM layer and then a dropout of

20% to reduce overfitting. We employed the Adam optimizer with a learning rate of $1e-3$ and employed categorical cross-entropy as the loss function. Lastly, we used fully connected dense layers. The last dense layer had two outputs for internal and external distraction detection, and the activation function was softmax.

2D CNN : The 2D Convolutional Neural Network has 2 convolution layers followed by a dropout (50%) and pooling layer (size=2), two fully connected layers, and a softmax layer respectively. Between two fully connected layers, the dropout layer (25%) is introduced as regularization in our model to reduce the over-fitting problem. The number of filters was 64 for the two CNN layers, with a kernel size of 5 and strides=2. We employed a learning rate of $1e-3$ with the Adam optimizer [21] and a loss function based on categorical cross-entropy. Throughout the training, the model went through a total of 50 epochs.

Time series raw data classification requires preprocessing to feed 2D CNN. Thus, [47] developed a method to convert time-series data into representative images that CNN can categorize by extracting spatial and temporal information. Then we can feed the 3 channels compound into a 2D CNN. Markov Transition Field (MTF) is the first image-generating technique, followed by Gramian Angular Summation Field (GASF) and Gramian Angular Difference Fields (GADF). MTF shows quantile transition probabilities. GADF encrypts time series. GASF is DADF with a sin difference instead of cos summation. Their Python library encoded our time series data. Since we selected 26 features, we converted 13,370 images, each 26x26 square size.

kNN: The k-nearest-neighbors (kNN) classifier is a straightforward and versatile algorithm that assigns a class label to a data point based on the class labels of its nearest neighbors in the training dataset. The value of k depends on the data. Low k values cause data variance overfitting (low training error, high test error). We tested k values from 1 to 10 and found that $k=5$ with the Minkowski parameter metric works best.

RF: To improve the performance of our RF model, we optimized our hyperparameters using the "RandomizedSearchCV" module from sklearn. To adapt the model for two-class classification, we adjusted the hyperparameters to estimator=100, max features='sqrt', and max depth=10. We then applied these hyperparameters to the model and presented our findings.

5 RESULTS

Figure 8 shows the pre-questionnaire mean ratings. We see that most subjects reported social distractions. We also summarized the post-questionnaire mean ratings in Figure 9. Participants' distractibility is indicated by pre- and post-questionnaire mean ratings. Most VR participants tried to focus, but they were distracted. Most participants enjoyed VR material and graphics/animations. We asked participants for comments or ideas (see appendix Table 9, Q12 to Q18) about the VR content, including what distracted and helped them learn. 19 of 27 participants found the audio and slides helpful, and 17 out of 27 found the animation component helpful. Only 9 participants found the avatar component helpful. Three people said the avatar distracted them, but most said all the components helped them learn. In addition, 22 of 27 participants said internal distraction tasks were harder than external ones. Participants also rated the distractors' distractibility. For internal distractions, Q2 (see Table 1) was ranked as the highest with a mean rank of 3.52, followed by Q1 at 3.07. The lowest ranks were Q3 with 1.63 and Q4 with 1.70. Similarly, for external distractors, the ranking from the highest to the lowest with mean rank was YouTube conversations (4.41) > ringtone (4.19) > door sound (2.52) > dialogues (2.07) > camera glitch effects (1.81).

We conducted cross-subject and cross-session tests to evaluate our models' performance for distraction detection tasks. Since there were 27 participants, we used 9-fold cross-validations for cross-subject testing, using the data from 24 subjects for training and the remaining three subjects' data for testing. Our results show that the RF model provides the best accuracy (over 83%) compared to the other models tested and the 2D CNN provides better results than the 1D CNN-LSTM model (see Table 4). The precision, recall, and F1-score for two-class classification from the cross-subject test are reported in Table 5.

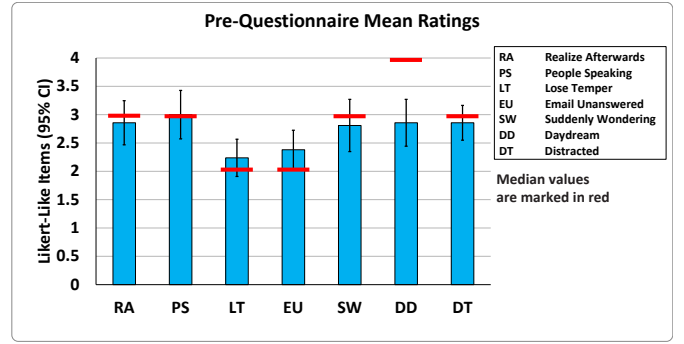


Fig. 8: Pre-questionnaire mean ratings.

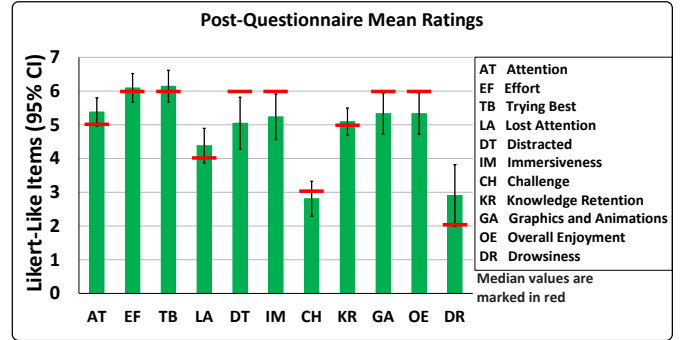


Fig. 9: Post-questionnaire mean ratings.

Table 4: Mean Accuracy of Cross-subject test from 1D CNN-LSTM, 2D CNN, kNN, and RF Models for classifying internal and external distraction.

Model Name	Accuracy with EEG features	Accuracy with Eye-gaze features	Accuracy with EEG plus eye-gaze features
1D CNN-LSTM	63.23%	69.40%	71.53%
2D CNN	63.08%	70.53%	73.15%
kNN	69.40%	76.06%	81.74%
RF	69.88%	77.31%	83.60%

We also perform cross-session testing for the generalizability of our model on new data. After training the model from three sessions, we tested its accuracy using data from the fourth session. As each session differed in duration and content, the test set varied in each case. The accuracy for each session as a test case, as provided by our models, is shown in Figure 10). We see that the best accuracy was over 79% and 77% provided by RF for session-I and session-IV, and the other sessions have slightly lower results. Thus, among the four models, the RF model performs best for most sessions with less computation time. Among the three feature sets, statistical features of the 1-second sliding window provide better results than image and raw features of time series data. Moreover, the computation time increases for the 2D CNN model since it extracts features from image data and it requires a reasonable time to tune hyper-parameters. Thus, we used RF for evaluating the precision, recall, and F1-scores for the individual class performance of cross-session, and the results are reported in Table 6.

According to our results (see Table 4), the RF model provided reasonably good results with less computation time, using a combination of EEG and ET data, for classifying internal and external distractions. To evaluate our best candidate model, the RF model, we collected data from a few people on a different day to see how it would perform. We collected data from two participants after 5 weeks using one session from the internal distraction phase (Phase I, session 1) and one from the external distraction phase (Phase II, session 2). We tested the RF model

Table 5: Mean Precision, recall, and F1-score of all models for classifying internal and external distraction from cross-subject test

Model Name	Class	precision %	recall %	F1-score %
1D CNN-LSTM	ID	0.62	0.68	0.66
	ED	0.72	0.77	0.74
2D CNN	ID	0.58	0.69	0.66
	ED	0.74	0.82	0.78
kNN	ID	0.87	0.74	0.81
	ED	0.76	0.90	0.82
RF	ID	0.87	0.77	0.82
	ED	0.78	0.91	0.84

Table 6: RF Model's precision, recall, and F1-score from cross-session test using three sessions for training and the remaining session for testing (test session indicated in column 1)

Session	Class	precision %	recall %	F1-score %
1	ID	0.69	0.74	0.73
	ED	0.78	0.84	0.81
2	ID	0.67	0.71	0.68
	ED	0.69	0.77	0.72
3	ID	0.68	0.60	0.65
	ED	0.67	0.75	0.69
4	ID	0.71	0.73	0.72
	ED	0.75	0.84	0.79

using the pre-trained model from the previous data, and the accuracy for the two participants was 79.55% and 77.62%.

Classification Accuracies for each Session

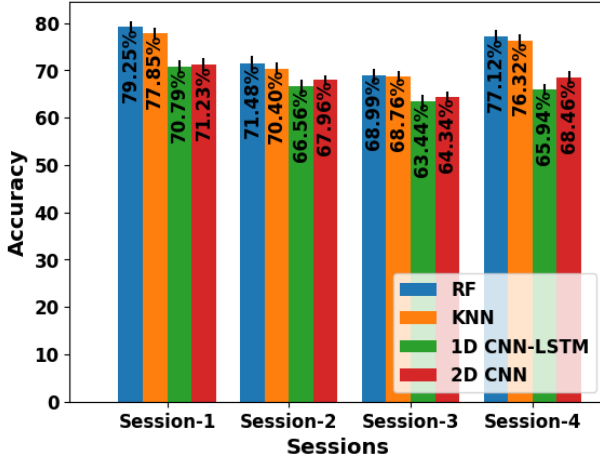


Fig. 10: Cross-session accuracy of ML models using 3 sessions for training and the remaining session for testing (indicated on the x-axis)

Our results are not far from the past non-VR research [45, 49] on attention/distraction classification (60% to 73% accuracy) using different fusion approaches of EEG and ET data. Thus, our results are not optimized for EVR applications. This EVR experiment may not have distracted all the participants equally, and obtaining high accuracy is challenging due to individual variability. Since the accuracy was not optimized, we wanted to see if gender-based grouping or personalized ML can improve performance.

To evaluate the impact of gender on the classification of internal and external distraction, we grouped data based on gender. Since our dataset contains ten female and seventeen male participants, we used an equal number of sample data from ten female and ten male. We employed five-fold cross-validation for the statistical features set. Table 7 presents the mean accuracy and F1-score obtained from the five-fold cross-validation process for each data modality (EEG, EYE) and

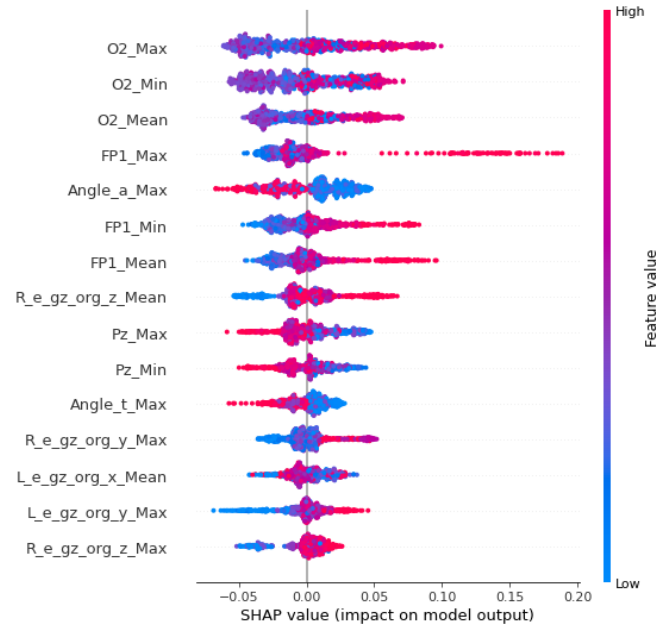


Fig. 11: SHAP analysis results using RF model. The red colony represents higher feature values and the blue colony represents lower feature values. The x-axis represents the feature contribution via the SHAP value for the specific prediction. A positive red SHAP value represents the impact of a prediction with external distraction when that feature value is high. Conversely, a negative red SHAP value represents the impact of a prediction with internal distraction when that feature value is high. Symbols in this figure: R -> right, L -> left, e -> eye, gz -> gaze, org -> origin, and x, y, z -> coordinates in a three-dimensional (x, y, z) gaze origin.

the combination of both data modalities. As shown in Table 7, the combination of EEG plus eye tracking data for the male group provides 90% accuracy and 89% F1-score. Similarly, combining both data for the female group provides 91% accuracy and 91% F1-score. Each data modality also demonstrated a significant increase in accuracy. In grouping, ET data offers better results than EEG data. The overall performance enhancements in the female group surpassed those in the male group across all modalities. The area under the ROC (Receiver Operating Characteristic) curve known as ROC-AUC for the RF model is shown in Figure 12 for different groups.

In addition, we also applied personalized ML using RF and kNN in which we split individual data (70:30 ratio) for training and testing. We found that both models provide similar results. The accuracy was around (86±10)%, and the individual accuracy for 20 of 27 participants was over 91%.

Post-hoc analysis was undertaken using the best-performing model (i.e. RF) in conjunction with the Shapely Additive Explanations (SHAP) method, which is a popular framework for interpreting the output of ML models [28]. To understand each feature's contribution to the model's output, the method allocates each feature a score depending on its significance in the model's prediction. The top 15 features (14% of the 104 features total) from the SHAP analysis are shown in Figure 11. The analysis revealed that gaze angle towards the avatar, gaze origin (left and right eye), and head rotation had the most significant contributions among the eye-tracking features, and O2, FP1, and Pz contributed most significantly to the EEG features. The gaze angle indicates where participants are directing their visual attention within the virtual environment, suggesting their focus on specific objects, characters, or events. Similarly, gaze origin and head rotation reflect not only visual exploration but also engagement and immersion within the VR.

Table 7: Mean Accuracy and F1-score for Gender-based Group. This table presents the results obtained from the five-fold Cross-Validation method.

Data Source	Accuracy Without Grouping	F1-score Without Grouping	Accuracy With Male Grouping	F1-score With Male Grouping	Accuracy With Female Grouping	F1-score With Female Grouping
EYE	0.79	0.79	0.79	0.80	0.82	0.81
EEG	0.69	0.71	0.70	0.71	0.73	0.74
EYE + EEG	0.88	0.87	0.90	0.89	0.91	0.91

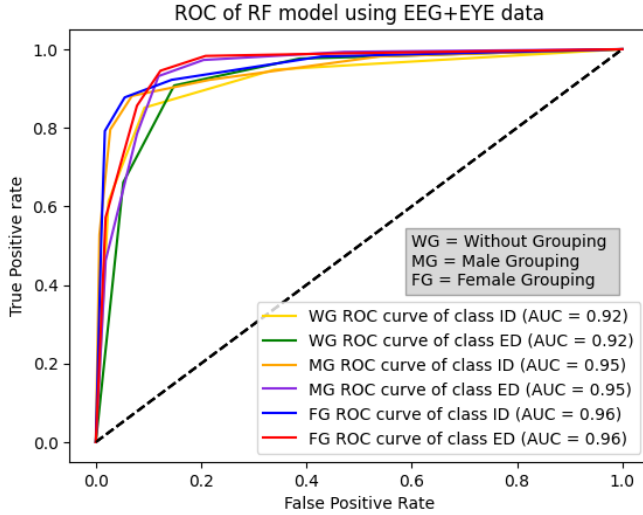


Fig. 12: The ROC-AUC curves for the different grouping with RF model using EEG plus eye gaze data.

6 DISCUSSION

We tested our models' ability to classify internal and external distractions using EEG and eye-gaze data. In an EVR setting, we classified students' internal and external distractions. Therefore, once the system detects the type of distraction a user is experiencing, it will alert the learner. We observe that RF performed slightly better than the other models tested with cross-subject, cross-session, and gender-based group tests. We also see that 2D CNN provides better results than 1D CNN, though 2D CNN requires more computation time than 1D CNN. The F1 scores with and without grouping and cross-subject testing are very close to each other, but they are much higher than those presented in cross-session testing. We believe that this lower F1-score for cross-session is due to the variability of session duration, the distance between the object to the user position, gaze angles, and the small sample size ($N = 27$). As each session differed in duration and content, the test set varied in each case. It might be possible that the window length for the (ID) class was interrupted by distracted sounds/tasks in between sessions. Our accuracy results with testing data from a different day were not that great (around 78%). In addition, we found that gender-based grouping significantly increases the accuracy of internal and external distraction detection. Our results indicate that the female group using EEG and ET data modalities collectively led to an accuracy value of 91% as shown in Table 7. To use gender-based models specific to a person's gender, they would be pre-trained and would need the user to choose the model that best fits them.

Our experiment has several drawbacks. While collecting data, participants go through the same experience (with different distractions) in both phases randomly. Thus, some participants might have potential biases after having already experienced the lectures. Since we did not investigate every possible internal and external distraction, our scope was limited. Additionally, some external distractions, such as single-person and two-person conversations, were similar in providing diverse insights. We may also have a gender bias in our results because we have 10 female and 17 male participants [35]. Furthermore, we con-

ducted our gender-based grouping result analysis using a small sample size. Our feature selection may have affected the classification models' generalization ability. We also acknowledge that we have investigated specific models, grouping, and early fusion approaches. However, numerous other models and potential data grouping including features, still need to be explored. Therefore, future research is necessary to develop a state-of-the-art neural network model to improve accuracy for real-time distraction detection. In addition, sharing eye-gaze data requires student privacy. We anonymized EEG and eye-tracking data from subjects who gave informed consent. However, because gaze data can reveal demographic information [27], it must be handled securely, especially when it involves school kids. To ensure security, regulatory compliance, gaze data misuse, and long-term storage must be dealt with carefully.

7 CONCLUSION AND FUTURE WORK

In this paper, we proposed a novel approach for classifying internal and external distractions. We developed an educational VR (EVR) environment with internal and external distractions intended to disrupt the participants' concentration while watching VR sessions. We collected EEG and eye-tracking data and applied 1D CNN-LSTM, 2D CNN, RF, and kNN models, for detecting internal and external distractions. We fused the data from the two modalities using an early feature fusion approach. Our cross-subject testing results show that the RF model had the highest accuracy of over 83% when multimodal data (EEG and eye-tracking combined) was pooled from all participants. When we performed cross-session testing, there was a significant drop in accuracy (ranging from 63% to 79%) for all the models. Additionally, gender-based grouping led to an accuracy value of 91% using the RF model. Thus, the RF model performed the best for the cross-subject, cross-session, and gender-based group testing. Furthermore, we also tested the RF model with the data collected on a different day, and the results were similar to when tested with new data (about 78%).

We believe similar methods could be used to classify distraction and no-distraction intervals in educational VR environments. However, we need further research to verify this, and would be our focus in the future. Additionally, we will investigate intermediate and late feature fusion along with state-of-the-art models for multimodal data and extract EEG features such as alpha, beta, and gamma waves to improve performance. The projected direction is to utilize these techniques to develop real-time distraction detection with more participants in various educational VR settings.

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A APPENDIX: PRE AND POST QUESTIONNAIRE QUESTIONS

Table 8: Pre-Questionnaire borrowed from [3]. Questions 1–7 were answered by participants using a 5-point Likert-style scale. Two questions (#8 and #9) asked for brief written responses.

Pre-Questionnaire Questions	
Q1	Do you say something and realize afterward that it might be taken as insulting?
Q2	Do you fail to hear people speaking to you when you are doing something else?
Q3	Do you lose your temper and regret it?
Q4	Do you leave important letters/emails unanswered for days?
Q5	Do you find yourself suddenly wondering whether you’ve used a word correctly?
Q6	Do you daydream when you ought to be listening to something?
Q7	Do you start doing one thing at home and get distracted into doing something else (unintentionally)?
Q8	Do you check your mobile in a regular classroom? If yes, how often, provide an approximate time interval like every 5 or 10 minutes?
Q9	What are the common distractions for you in a regular classroom?

Table 9: Post-Questionnaire adapted and modified from [3]. Questions 1-11 were answered using a 7-point Likert-type scale, and the responses were compiled. The questions from 12–15 were multiple-choice.

Post-Questionnaire Questions	
Q1	To what extent did the VR class hold your attention?
Q2	How much effort did you put into attending the VR class and quiz?
Q3	Did you feel you were trying your best?
Q4	To what extent did you lose attention?
Q5	Did you feel the urge to see what was happening around you?
Q6	To what extent you enjoyed the VR class and quiz exam, rather than something you were just doing?
Q7	To what extent did you find the VR class challenging?
Q8	How much knowledge you could retain after the VR class on solar panels?
Q9	To what extent did you enjoy the graphics and the animation?
Q10	How much would you say you enjoyed the VR class?
Q11	To what extent did you feel drowsiness?
Q12	Which one helped you to understand the lessons? a) audio b) slides c) avatar d) animations
Q13	Which one helped you to recall information to answer quizzes? a) audio b) slides c) avatar d) animations
Q14	Which component(s) distracted you except our simulated distractions? a) audio b) slides c) avatar d) animations
Q15	Did you feel any other distraction during VR class except our created distraction? a) Mind Wandering b) Internal Stress c) Others
Q16	Which event created more distraction during VR class?). a) Internal events. b) External events. c) Equal.
Q17	Rank internal distraction events (Highest(4) to lowest (1) distractors). a) Counting words started with "S". b) Counting avatar hand movements. c) Think about last weekend’s activities. d) Think about next weekend’s activities.
Q18	Rank external distraction events (Highest(5) to lowest (1) distractors). a) Mobile ringtone. b) Dialogue sounds. c) Door sound. d) YouTube conversations. e) Camera glitch effects